Arav Palsule

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LinkedIn — GitHub

B.E. Data Science (2027) — CGPA: 8.34/10

Professional Summary

Aspiring AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, and Generative AI. Skilled in Python, SQL, data preprocessing, feature engineering, predictive modeling, and model evaluation. Experienced in building LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, multi-agent workflows using LangGraph, and AI automation solutions. Familiar with context engineering, prompt engineering, embeddings, and API integrations.

Education

A. P. Shah Institute of Technology, Thane
Bachelor of Engineering in Data Science — 2023 – 2027
CGPA: 8.34/10

Technical Skills

Programming Languages: Python, SQL, JavaScript
Machine Learning: Supervised Learning, Predictive Modeling, Feature Engineering, Data Preprocessing, Model Training, Model Evaluation, Statistical Analysis
Artificial Intelligence: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Transfer Learning, Embeddings
Generative AI and Agentic AI: Large Language Models (LLMs), RAG, Vector Search, AI Agents, Multi-Agent Systems, LangGraph, Prompt Engineering, Context Engineering, Claude Skills
Libraries and Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Flask
Databases and APIs: MySQL, SQLite, Supabase, REST APIs, API Integration
Tools and Platforms: Git, GitHub, Zapier, Zoho

Projects

Radiator Routes – AI-Powered Travel Planning Platform
Developed an AI-powered travel planning platform featuring conversational assistance, personalized itinerary generation, travel memory, and dynamic replanning. Implemented multi-agent orchestration using LangGraph to coordinate specialized travel planning workflows and AI-assisted itinerary generation. Built deterministic itinerary validation covering budget constraints, scheduling conflicts, travel feasibility, geographic validity, and opening hours, with an automated LLM-based repair mechanism. Implemented group preference optimization using Least Misery scoring to compare itinerary alternatives and balance traveller preferences.

CrackX – AI-Powered Road Damage Detection System
Developed a road condition monitoring platform integrating YOLO-based computer vision to detect and localize road damage, including potholes and cracks. Implemented image and video processing workflows with bounding-box localization and damage severity assessment for road condition reporting. Integrated the detection pipeline into a management platform supporting road damage reporting and maintenance workflows.

Dengue Outbreak Prediction – Machine Learning
Developed a dengue case forecasting pipeline using historical case reports, weather variables, geographic attributes, and temporal features. Trained and compared four regression algorithms: Random Forest, Extra Trees, HistGradientBoosting, and Ridge Regression using log-transformed target values. Implemented chronological holdout and rolling-window validation using MAE, RMSE, and SMAPE to evaluate forecasting performance.

CineRec AI – AI-Powered Video Highlight Generator
Developed an AI-powered video processing pipeline combining scene emotion classification, scene importance ranking, and semantic music matching. Integrated NLP and Deep Learning components, including DistilBERT-based emotion classification and embedding-based semantic music retrieval. Automated timestamp generation, video clip extraction, soundtrack alignment, and final reel composition using Python, FFmpeg, and MoviePy.

Experience

AI Intern – Ennoble Social Innovation September 2026 – October 2026

- Created custom Claude Skills using structured context engineering to automate business tasks, including PDF formatting according to internal document structures and standards.

- Developed an employee focus-monitoring prototype integrating face detection and emotion analysis to estimate focus scores and identify potential sleep or inactivity events.
- Designed automated screenshot capture and HR reporting workflows to communicate detected events and employee focus patterns.
- Built AI-powered business process automations using Zapier, LLMs, and API integrations to streamline repetitive operational tasks.

Achievements

- **Rank 10** – C.R. Rao Statistics & Probability Olympiad: Ranked among 2.5 lakh+ participants across Asia.
- **AIR 3192, AICR 35** – Aryabhatta National Maths Competition 2026: Scored 100% in Round 1 and 80% in Round 2.
- **2nd Runner-Up** – QUASAR 4.0 Hackathon: Delivered a technical solution under a 24-hour development deadline.
- **Elite Rank** – NPTEL Multivariable Calculus: Recognized for mathematical and quantitative problem-solving skills.

Leadership

Founding President – Maths & Applied Science Club — 2026

- Founded and led the club, organizing 25+ technical events to promote mathematics, science, and technical learning among students.

Research Member – DSA Club — 2025 – 2026

- Explored LLMs, Retrieval-Augmented Generation (RAG), and machine learning pipelines; mentored junior members in technical concepts.